

Steam Reviews Dataset: A Big Data Analysis by Ali Daoud

Abstract

This project presents a Big Data analysis of the Steam Reviews Dataset using Apache Hadoop and Apache Spark. The dataset was obtained from Kaggle and the analyzed file contained approximately 21.7 million user reviews from the Steam gaming platform, with a total size of approximately 7.6 GB. Due to its large volume, the dataset was suitable for large-scale data analysis.

The dataset was stored in the Hadoop Distributed File System (HDFS), while the analysis was performed using PySpark. Preprocessing included schema inspection and conversion of selected columns from string values into numerical formats to enable statistical calculations. Descriptive and exploratory analyses were then conducted to identify patterns related to player recommendations, game popularity, language distribution, purchase behavior, and player engagement.

The results showed that positive reviews greatly outnumbered negative reviews. English was the most common review language, and several major multiplayer games were among the most reviewed games in the dataset. Additional analyses examined purchase behavior, free copies, helpful votes, and player playtime in relation to recommendation status. The project demonstrates how Hadoop can be used for large-scale data storage and Spark for processing and analyzing large datasets.

Content

Steam Reviews Dataset: A Big Data Analysis by Ali Daoud.....	1
1.1 Dataset.....	2
1.2 Introduction to the Tools	3
1.2.1 Hadoop.....	3
1.2.2 Spark	3
2. Materials and Methods	3
2.1 Hardware	3
2.2 Data Collection	3
2.3 Data Pre-processing.....	3
2.4 Data Analysis, Representation and Visualization.....	4
2.5 Security	4
3. Results	4

4. Discussion	7
5. Conclusion	8
6. References	8
7. Appendix.....	8

1. Introduction

The rapid growth of digital gaming platforms has generated massive amounts of user-generated content in the form of online reviews. These reviews provide valuable information about player satisfaction, game quality, and consumer behavior. As gaming platforms continue to expand, the volume of available review data becomes increasingly difficult to process using traditional analytical methods. Big Data Analytics offers an effective approach for handling large-scale datasets and extracting meaningful insights from millions of records.

This project focuses on the Steam Reviews Dataset obtained from Kaggle. The dataset contains approximately 21.7 million game reviews and occupies approximately 7.6 GB of storage, making it suitable for Big Data Analytics. The dataset includes information about games, user recommendations, languages, purchase status, playtime statistics, and review metadata. Due to its size and complexity, the dataset represents a realistic big data scenario requiring distributed storage and processing technologies.

The main objective of this project is to analyze player reviews and identify patterns related to game popularity, recommendation behavior, language distribution, purchase behavior, and player engagement. Hadoop Distributed File System (HDFS) was used for distributed storage of the dataset, while Apache Spark and PySpark were used for large-scale data processing and analysis. The project demonstrates how modern big data technologies can be applied to extract valuable insights from large gaming datasets.

1.1 Dataset

The dataset used in this project is the Steam Reviews Dataset downloaded from Kaggle. The dataset contains approximately 21.7 million reviews collected from the Steam gaming platform. Each record contains information such as game identifier, game name, review text, review language, recommendation status, helpful votes, purchase information, and player activity statistics. The raw dataset size is approximately 7.6 GB and therefore qualifies as a big data dataset according to the requirements of the project.

Dataset Summary

Metric	Value
Total reviews	21,747,371
Positive reviews	19,022,610
Negative reviews	2,724,761
Number of games	315
HDFS dataset size	7.6 GB

1.2 Introduction to the Tools

1.2.1 Hadoop

Apache Hadoop was used in this project as the storage platform for the Steam Reviews Dataset. Since the dataset contains over 21.7 million reviews and occupies approximately 7.6 GB, storing and managing the data in a distributed environment was appropriate. The dataset was uploaded to HDFS, where it was stored as distributed blocks for storage and management purposes. This setup enabled reliable handling of large amounts of data and supported scalable processing throughout the project.

1.2.2 Spark

Apache Spark was used as the main processing framework for analyzing the Steam Reviews Dataset. Through PySpark, the dataset was loaded into Spark DataFrames and processed using distributed computations. Spark DataFrame operations were used for grouping, counting, aggregation, sorting, and statistical calculations on millions of records. The framework was used to generate the results presented in this report, including analyses of recommendation behavior, language distribution, game popularity, purchase patterns, free-copy status, helpful votes, and player engagement.

2. Materials and Methods

2.1 Hardware

Table 1. Hardware specifications used in the project.

Component	Specification
CPU	Intel Core i5-1135G7 (11th Gen, 4 Cores, 8 Threads)
RAM	8 GB
Storage	SSD Storage
Operating System	Windows 11 Enterprise Hadoop Version
Hadoop	3.3.6
Spark Version	Spark 3.5.1

2.2 Data Collection

The Steam Reviews Dataset was downloaded from Kaggle and uploaded to the Hadoop Distributed File System (HDFS) using Hadoop commands, where it was stored in the directory `"/data/steam_reviews.csv"`. After verifying that the dataset had been successfully stored in HDFS, the dataset was loaded locally into Apache Spark using PySpark for analysis.

2.3 Data Pre-processing

The dataset was initially loaded into PySpark DataFrames using Spark's CSV reader with header support enabled. During the preprocessing phase, the structure of the dataset was inspected through schema analysis. Several columns containing numerical information were initially stored as string values and therefore required type conversion before statistical operations could be performed.

Selected columns, including `votes_helpful` and `author.playtime_forever`, were converted from string format to numeric data types to enable aggregation functions and statistical calculations. This preprocessing stage prepared the dataset for further analysis.

2.4 Data Analysis, Representation and Visualization

The analysis focused on identifying patterns in user recommendations, game popularity, language distribution, purchase behavior, and player engagement. Descriptive statistical techniques were applied using Spark DataFrame operations such as grouping, counting, aggregation, and sorting.

Several analytical queries were executed to determine the distribution of positive and negative recommendations, identify the most reviewed games, analyze language usage among players, examine purchase behavior and free copies, and compare average helpful votes and average playtime between positive and negative reviews. The results were presented using tables and bar charts to facilitate interpretation and communication of the findings.

2.5 Security

HDFS file permissions were used to restrict access to the dataset. The permissions for `/data/steam_reviews.csv` were configured to 600, which grants read and write access only to the file owner while denying access to group members and other users. This provided a basic access-control mechanism for protecting the dataset stored in HDFS.

3. Results

The Steam Reviews Dataset contains approximately 21.7 million reviews and provides valuable insights into player behavior and game popularity. Figure 1 presents the distribution of recommendations among users. The results show that positive recommendations significantly outnumber negative recommendations, indicating that most players were satisfied with their gaming experiences.

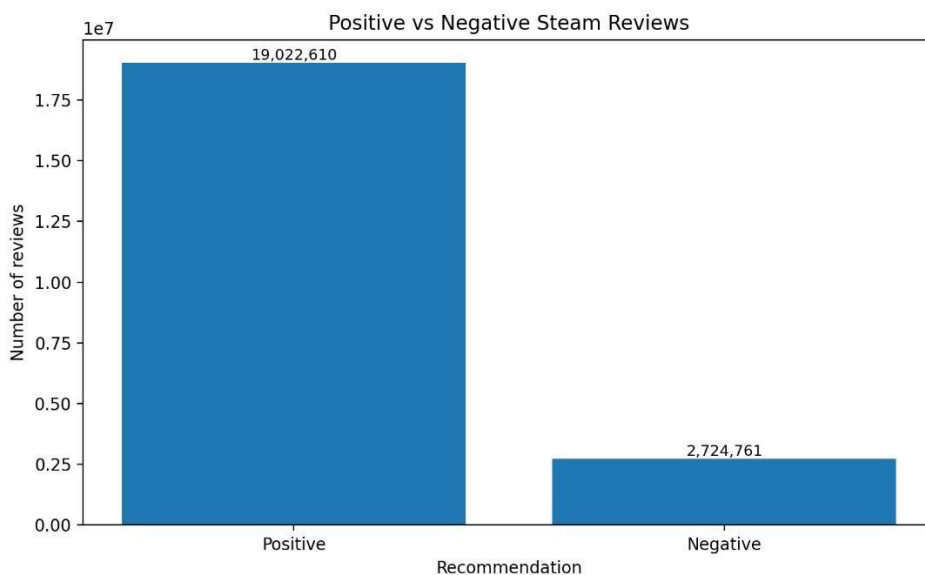


Figure 1. Positive vs Negative Steam Reviews.

Figure 2 presents the ten most reviewed games in the dataset. PLAYERUNKNOWN'S BATTLEGROUNDS received the largest number of reviews, followed by Grand Theft Auto V and Tom Clancy's Rainbow Six Siege. These results highlight the popularity and engagement levels of major multiplayer games.

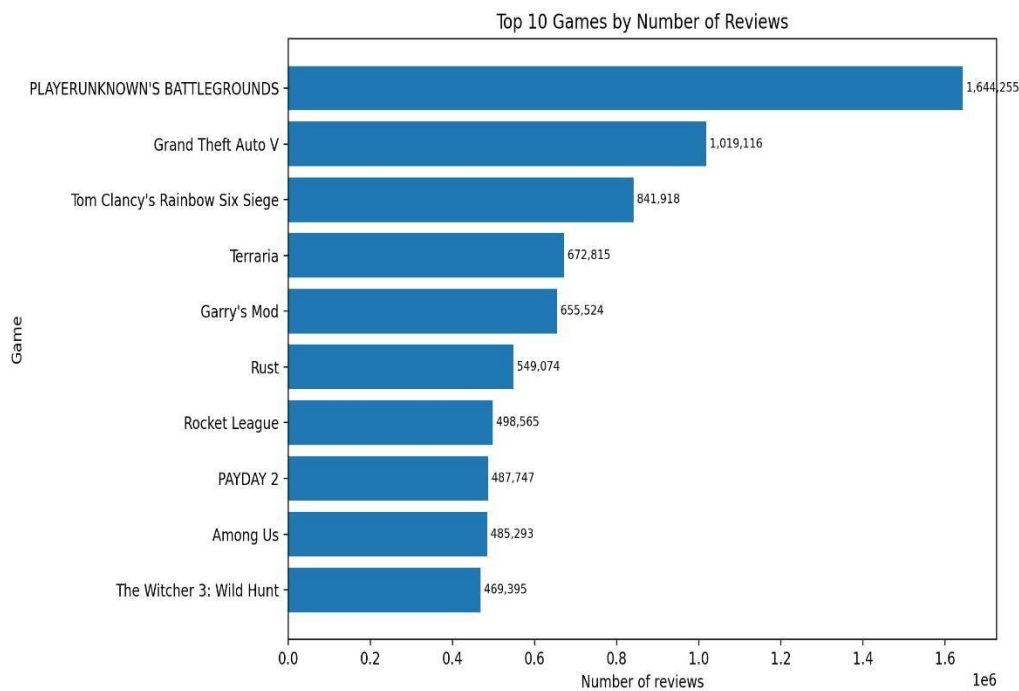


Figure 2. Top 10 Games by Number of Reviews.

Figure 3 illustrates the distribution of review languages. English was by far the most frequently used language, followed by Simplified Chinese and Russian. This reflects the global nature of the Steam platform and its diverse user base.

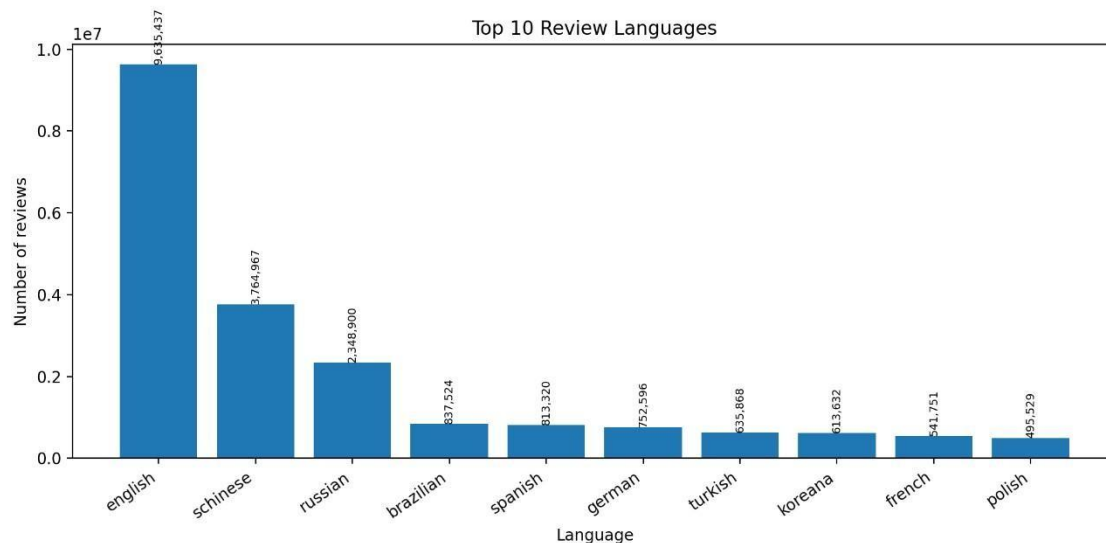


Figure 3. Distribution of Review Languages.

Figure 4 presents the distribution of reviews by Steam purchase status. Most reviews were submitted by users who had purchased the game through

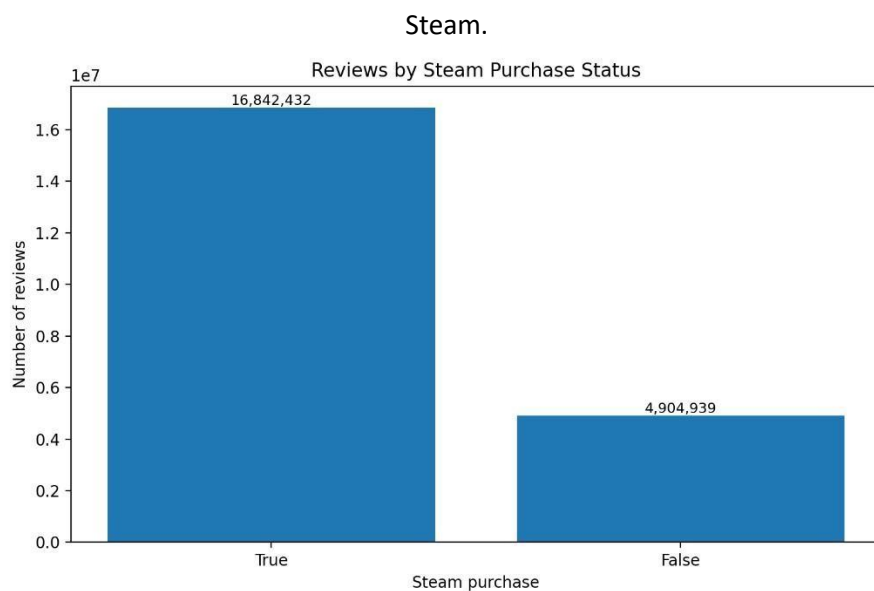


Figure 4. Reviews by Steam Purchase Status.

Figure 5 presents the distribution of reviews by free-copy status. The majority of reviews were written by users who did not receive the game for free.

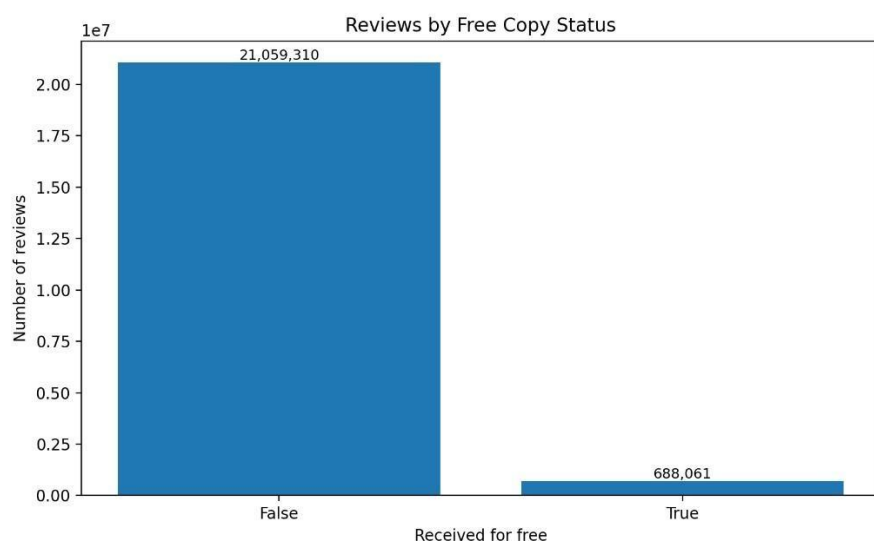


Figure 5. Reviews by Free Copy Status.

Figure 6 presents the average number of helpful votes for positive and negative reviews. Negative reviews received a higher average number of helpful votes compared to positive reviews, suggesting that critical reviews may contain more detailed information that other users find useful.

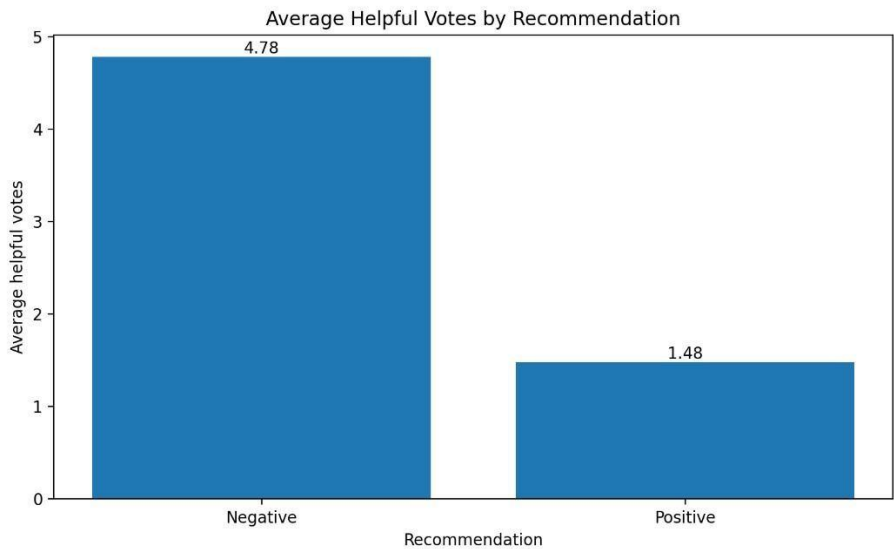


Figure 6. Average Helpful Votes by Recommendation.

Figure 7 compares average playtime between positive and negative reviewers. Players who submitted negative reviews had, on average, more playtime than players who submitted positive reviews. This may indicate that experienced players are more likely to identify issues and provide critical feedback.

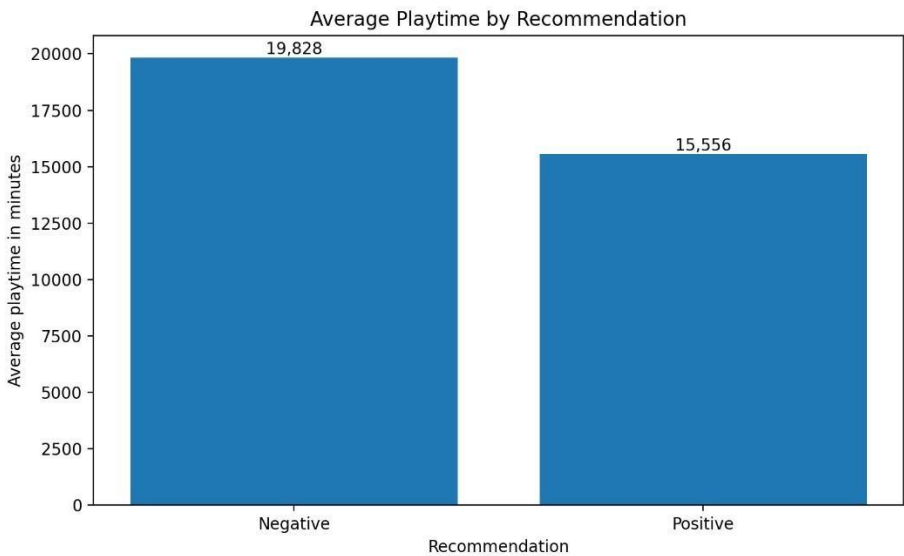


Figure 7. Average Playtime by Recommendation.

Table 2 presents the relationship between recommendation status, Steam purchase status, and whether the game was received for free. Positive recommendations were more common across all categories.

Recommendation Relationships

Category	Count	on ew
Steam purchase + Positive	14,731,705	
Steam purchase + Negative	2,110,727	
No Steam purchase + Positive	4,290,905	
No Steam purchase + Negative	614,034	
Not free + Positive	18,411,577	
Not free + Negative	2,647,733	
Free + Positive	611,033	
Free + Negative	77,028	

Table 2. Recommendation Relationships by Purchase and Free-Copy Status.

The popularity analysis showed that several major multiplayer and competitive games were among the most reviewed games in the dataset. Furthermore, negative reviews received a higher average number of helpful votes than positive reviews. This may suggest that users found negative reviews particularly useful when evaluating games.

The playtime analysis showed that users who submitted negative reviews had, on average, more playtime than users who submitted positive reviews. This may indicate that players with more gameplay experience are more likely to identify problems or express critical opinions. Overall, the findings demonstrate how large-scale gaming datasets can provide meaningful insights into player behavior and review patterns.

5. Conclusion

This project demonstrated the use of Hadoop and Spark technologies for large-scale data analytics using the Steam Reviews Dataset. The dataset was stored in HDFS, while PySpark was used to process and analyze approximately 21.7 million reviews. Descriptive analyses identified patterns related to recommendation behavior, game popularity, language distribution, purchase status, free-copy status, helpful votes, and player playtime.

The results demonstrate how Big Data technologies can be applied to large gaming datasets to identify meaningful patterns in user behavior. HDFS provided a suitable solution for storing the large dataset, while Spark and PySpark enabled efficient analysis of millions of records. Overall, the project illustrates how large-scale data processing can support the exploration and interpretation of user-generated gaming data.

6. References

Najzeko. *Steam Reviews Dataset 2021* [Internet]. Kaggle; 2021 [cited 2026 Jun 8]. Available from: <https://www.kaggle.com/datasets/najzeko/steam-reviews-2021>.

Apache Software Foundation. *Apache Hadoop Documentation* [Internet]. [cited 2026 Jun 8]. Available from: <https://hadoop.apache.org/docs/>.

Apache Software Foundation. *Apache Spark Documentation* [Internet]. [cited 2026 Jun 8]. Available from: <https://spark.apache.org/docs/latest/>.

7.Appendix

<https://github.com/fetyandaoud/DA381A-BigData-Project>

Requirement Implemented Ny Comment		
Req. 1: Data Collection	Yes	Steam Reviews Dataset (approximately 21.7 million reviews, 7.6 GB) was obtained from Kaggle and stored in HDFS.

Requirement Implemented Ny Comment

Req. 2: Data Preprocessing	Yes	Dataset preprocessing in PySpark included schema inspection and conversion of selected columns from string format to numeric data types.
Req. 3: Smart Data Analysis	Yes	Descriptive and exploratory analyses were performed to identify patterns related to recommendations, game popularity, language distribution, purchase behavior, and player engagement.
Req. 4: Representation & Visualization	Yes	Results were presented through tables and bar charts based on Spark analysis outputs.
Req. 5: Security	Yes	HDFS file permissions were configured to restrict access to the dataset. Only the file owner was granted read and write permissions.